

Université Internationale de Rabat

THE INNOVATIVE UNIVERSITY

Hands-on on Quantum Computing

Deutsch-Jozsa Algorithm

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Lab's objective

This lab introduces the Deutsch-Jozsa algorithm (DJA), while showing how a quantum computer can determine if a function is constant or balanced. We will explore superposition and interference, and see how quantum parallelism allows evaluating all inputs simultaneously while showing how faster than the classical method.

I Classical algorithm

Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function, which is considered either :

- **Constant :** $f(x)$ has the same value for all inputs x.
- **Balanced :** $f(x)$ outputs 0 for exactly half of the inputs and 1 for the other half.

The goal is to determine whether f is constant or balanced. Since f is given as a black box, the only way to determine its type is by querying the function on different inputs. So, we proceed by evaluating $f(x)$ for different inputs until eliminating one of the two possibilities.

Algorithm 1 Classical DJA

Require: Black-box function $f : \{0,1\}^n \to \{0,1\}$
Ensure: Determine whether f is Constant or Balanced

```
Evaluate v ← f(0^n)
for each x ∈ {0,1}^n with x ≠ 0^n do
    Evaluate f(x)
    if f(x) ≠ v then
        return Balanced
    end if
end for
return Constant
```

As explained in the abovementioned algorithm, we first evaluate the function at the input $0^n : v = f(0^n)$. This gives us a reference output. We then evaluate $f(x)$ for other inputs x. If at any point we find : $f(x) \neq v$, then the function must be **Balanced**, since a constant function would produce the same output for every input.

In the worst case, the first 2^{n-1} evaluations all return the same value. Balanced functions must have exactly 2^{n-1} zeros and 2^{n-1} ones. Therefore, if we observe $2^{n-1} + 1$ identical outputs, the function cannot be balanced and must be constant. Thus, the classical query complexity is : $O(2^n)$, which grows exponentially with n.

II Quantum algorithm using one input qubit

This section presents the mathematical analysis of Deutsch's algorithm using one input qubit and one ancilla qubit. Its objective is to determine whether a Boolean function $f : \{0,1\} \to \{0,1\}$ is *constant*, meaning $f(0) = f(1)$, or *balanced*, meaning $f(0) \neq f(1)$. The implementation considers all four possible functions.

Oracle type	Function	$f(0)$	$f(1)$	Class
`constant_0`	$f(x) = 0$	0	0	Constant
`constant_1`	$f(x) = 1$	1	1	Constant
`balanced_identity`	$f(x) = x$	0	1	Balanced
`balanced_not`	$f(x) = 1 \oplus x$	1	0	Balanced

The quantum oracle is defined by

$$U_f \left|x\right\rangle \left|y\right\rangle = \left|x\right\rangle \left|y \oplus f(x)\right\rangle \tag{1}$$

where $\oplus$ denotes XOR. Throughout, two-qubit kets use the logical order $\left|x\right\rangle \left|y\right\rangle$ (input followed by ancilla).

Step 1 : Initial State
Both qubits are initialized to $\left|0\right\rangle$:

$$\left|\psi_0\right\rangle = \left|0\right\rangle \left|0\right\rangle . \tag{2}$$

Step 2 : Preparing the Ancilla
A Pauli-X gate followed by a Hadamard gate prepares

$$HX \left|0\right\rangle = H \left|1\right\rangle = \frac{\left|0\right\rangle - \left|1\right\rangle}{\sqrt{2}} = \left|-\right\rangle . \tag{3}$$

The combined state becomes

$$\left|\psi_1\right\rangle = \left|0\right\rangle \otimes \frac{\left|0\right\rangle - \left|1\right\rangle}{\sqrt{2}}. \tag{4}$$

Step 3 : Superposition of the Input
Apply a Hadamard gate to the input :

$$H \left|0\right\rangle = \frac{\left|0\right\rangle + \left|1\right\rangle}{\sqrt{2}} = \left|+\right\rangle . \tag{5}$$

Consequently,

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \tag{6}$$

Expanding in logical order,

$$|\psi_2\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle). \tag{7}$$

Step 4 : Oracle and Phase Kickback
Because the ancilla is $|-\rangle$, the oracle satisfies

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle . \tag{8}$$

Acting on the input superposition gives

$$|\psi_3\rangle = \frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \otimes |-\rangle . \tag{9}$$

Case A : ***constant_0****,* $f(x) = 0$
Here $f(0) = f(1) = 0$, so both phase factors equal $+1$:

$$|\psi_3\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |-\rangle = |+\rangle |-\rangle . \tag{10}$$

Case B : ***constant_1****,* $f(x) = 1$
Here $f(0) = f(1) = 1$, so both phase factors equal -1 :

$$|\psi_3\rangle = -\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |-\rangle = - |+\rangle |-\rangle . \tag{11}$$

The overall minus sign is a global phase and does not affect measurement probabilities.
Case C : ***balanced_identity****,* $f(x) = x$
Here $f(0) = 0$ and $f(1) = 1$:

$$|\psi_3\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \otimes |-\rangle = |-\rangle |-\rangle . \tag{12}$$

Case D : ***balanced_not****,* $f(x) = 1 \oplus x$
Here $f(0) = 1$ and $f(1) = 0$:

$$|\psi_3\rangle = \frac{- |0\rangle + |1\rangle}{\sqrt{2}} \otimes |-\rangle = - |-\rangle |-\rangle . \tag{13}$$

For the balanced oracles, the relative phase between input components differs from that of the constant oracles. The ancilla remains in $|-\rangle$ up to a global phase of the joint state.

Step 5 : Quantum Interference
Apply a final Hadamard gate to the input qubit. Since $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$, the four outcomes are :

$$\texttt{constant_0}: \quad |\psi_4\rangle = |0\rangle|-\rangle, \tag{14}$$
$$\texttt{constant_1}: \quad |\psi_4\rangle = -|0\rangle|-\rangle, \tag{15}$$
$$\texttt{balanced_identity}: \quad |\psi_4\rangle = |1\rangle|-\rangle, \tag{16}$$
$$\texttt{balanced_not}: \quad |\psi_4\rangle = -|1\rangle|-\rangle. \tag{17}$$

The final Hadamard gate converts relative phase information into a computational-basis measurement outcome. Global phases have no effect on the observed result.

Step 6 : Measurement Only the input qubit is measured. For either constant oracle,

$$P(0) = 1, \qquad P(1) = 0. \tag{18}$$

For either balanced oracle,

$$P(0) = 0, \qquad P(1) = 1. \tag{19}$$

Oracle	Final input state	Measurement
`constant_0`	$\|0\rangle$	0
`constant_1`	$-\|0\rangle$	0
`balanced_identity`	$\|1\rangle$	1
`balanced_not`	$-\|1\rangle$	1

General Mathematical Result
For any Boolean function $f : \{0,1\} \to \{0,1\}$, the final input state is

$$|\phi\rangle = \frac{1}{2}\Big[\big((-1)^{f(0)} + (-1)^{f(1)}\big)|0\rangle + \big((-1)^{f(0)} - (-1)^{f(1)}\big)|1\rangle\Big]. \tag{20}$$

Hence,

$$P(0) = \frac{1}{4}\left[(-1)^{f(0)} + (-1)^{f(1)}\right]^2, \tag{21}$$
$$P(1) = \frac{1}{4}\left[(-1)^{f(0)} - (-1)^{f(1)}\right]^2. \tag{22}$$

Therefore,

$$\text{Measurement} = \begin{cases} 0, & \text{if } f \text{ is constant,} \\ 1, & \text{if } f \text{ is balanced.} \end{cases} \tag{23}$$

Remarks
Deutsch's algorithm uses superposition, phase kickback, and interference to determine whether $f(0) = f(1)$ with one oracle call. Although the four functions produce different phases, both constant functions yield measurement 0 and both balanced functions yield measurement 1, without separately measuring the two function values.

III Quantum algorithm using two two qubit

Let us now apply it for $n = 2$. So, we use 2 input qubits, 1 ancilla qubit and 2 classical bits. All qubits start in $|000\rangle$. To prepare Ancilla, we apply $X \to H$ to the ancilla qubit :

$$|0\rangle \xrightarrow{X} |1\rangle \xrightarrow{H} \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

The global state becomes :

$$|\psi_0\rangle = |00\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Next step focuses on creating the superposition by applying Hadamard gates to input qubits :

$$H^{\otimes 2} |00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

This creates quantum parallelism, so the system becomes :

$$|\psi_1\rangle = \frac{1}{2} \sum_{x \in \{0,1\}^2} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

After creating the superposition, Oracle could be applied. So, the oracle implements :

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

Because the ancilla is in

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

the oracle produces phase kickback :

$$U_f \left(|x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Thus the global state becomes :

$$|\psi_2\rangle = \frac{1}{2} \sum_x (-1)^{f(x)} |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Applying Hadamard gates again (Interference) :

$$H^{\otimes 2} \sum_x (-1)^{f(x)} |x\rangle$$

Key result :

$$\text{Amplitude of } |00\rangle = \frac{1}{4} \sum_x (-1)^{f(x)}$$

So if f is Constant,

$$\sum_x (-1)^{f(x)} = \pm 4$$

The Amplitude of $|00\rangle$ is non-zero and the measurement result is $\boxed{00}$
If f is Balanced,

$$\sum_x (-1)^{f(x)} = 0$$

Amplitude of $|00\rangle$ is zero and the measurement result is $\boxed{\text{Anything except 00}}$

Even though DJA does not have significant practical use, it highlights well the foundation for many other quantum algorithms, notably by introducing the quantum interference effect. Two of the most important algorithms in the field, Shor's algorithm and Grover's algorithm, were inspired by its working principles.